TEXORA · LECTURE SHEET

Noise Pollution

Sound and the decibel, how noise levels add and fall with distance, where industrial noise comes from, how exposure is evaluated against standards, and control at source, path and receiver.

TE-4204 · ENVIRONMENTAL ENGINEERING AND POLLUTION CONTROL · TOPIC 5 · LEVEL 4 TERM 2

1. Sound, noise and the decibel

Sound is a pressure wave in air produced by a vibrating source; **noise** is simply unwanted sound. Two properties matter: **frequency** (pitch, in hertz — the human ear hears roughly 20 Hz to 20 kHz) and **intensity** (energy, related to pressure). The range of pressures the ear handles is enormous, so sound level is measured on a compressed logarithmic scale, the **decibel (dB)**:

$$\text{Sound level (dB)} = 10 \times \log_{10} (I \div I_0)$$

where I_0 is the reference intensity at the threshold of hearing. Consequences students must internalise: the scale is logarithmic, so levels cannot be added arithmetically; a 10 dB increase means ten times the intensity and sounds roughly twice as loud; and a 3 dB increase means double the energy — the basis of exposure rules that halve permitted time for each 3 dB rise. Measurements for environmental and occupational purposes use the **A-weighted** scale, dB(A), which filters sound roughly as the ear does.

2. Adding noise levels — worked examples

For independent (uncorrelated) sources, intensities add; levels do not:

$$L_{total} = 10 \times \log_{10} (\Sigma \, 10^{L_i/10})$$

WORKED EXAMPLE A — COMBINING MACHINE NOISE LEVELS

Two identical looms each produce 88 dB(A) at an operator's position.

$L = 10 \times \log_{10}(10^{8.8} + 10^{8.8}) = 10 \times \log_{10}(2 \times 10^{8.8}) = 88 + 10 \times \log_{10} 2 = 88 + 3.0 =$ **91 dB(A)**.

Rule: doubling the number of equal sources adds only 3 dB — but 3 dB is double the sound energy, so exposure dose doubles too.

Three different sources measure 90, 88 and 84 dB(A):

Sum of intensities $\propto 10^{9.0} + 10^{8.8} + 10^{8.4} = 1.00\times10^{9} + 0.63\times10^{9} + 0.25\times10^{9} = 1.88\times10^{9}$

$L_{total} = 10 \times \log_{10}(1.88 \times 10^{9})$ = **92.7 dB(A)**.

Note how the loudest source dominates: removing the 84 dB source entirely would reduce the total only to about 92.4 dB. Control effort belongs on the loudest source first.

WORKED EXAMPLE B — NOISE FALLS WITH DISTANCE

For a point source in the open, level falls by 6 dB at each doubling of distance:

$$L_2 = L_1 - 20 \times \log_{10}(r_2 \div r_1)$$

A generator measures 95 dB(A) at 1 m. At 8 m: $L = 95 - 20 \times \log_{10} 8 = 95 - 18.1$ = **76.9 dB(A)**. Inside a reverberant shed the fall is less than this free-field value, because reflected sound adds to the direct sound — one reason shed treatment (Section 5) matters.

3. Industrial noise sources and their evaluation

SOURCE	NOISE CHARACTER	EVALUATION NOTES
Weaving shed (many looms)	Continuous, broadband, impact peaks at each pick	Whole-hall level matters; many equal sources add by the 3 dB-per-doubling rule
Spinning / knitting halls	Continuous machinery hum and whine	High-frequency components carry far; ear protection zones common
Generators, compressors, boilers	Continuous low-frequency rumble + exhaust roar	Low frequency travels and penetrates structures; enclosures must be sealed
Pumps, fans, stenters	Continuous, tied to rotational speed	Often the neighbourhood complaint source at night
Intermittent: material handling, workshops	Impulsive peaks	Peak levels, not just averages, must be assessed

Evaluation proceeds in steps: (1) survey with a **sound level meter** — readings at operator positions and at the boundary, recording the equivalent continuous level L_{eq} over a representative period; (2) personal **dosimetry** for mobile workers, giving their true shift exposure; (3) **octave-band analysis** where control is being designed, because low and high frequencies need different treatments; (4) comparison with the applicable limits. Two families of limits apply: **occupational** limits inside the workplace — 85 dB(A) averaged over 8 hours is the widely adopted action criterion internationally, with permitted time halving per 3 dB rise — and **environmental** limits at the boundary, which in Bangladesh are set zone-wise under the Environment Conservation Rules:

ZONE	DAY (DB(A))	NIGHT (DB(A))
Silent zone (hospitals, schools)	50	40
Residential	55	45
Mixed (residential + commercial/industrial)	60	50
Commercial	70	60
Industrial	75	70

Effects to be evaluated are not only hearing damage (**noise-induced hearing loss** from chronic exposure, and acoustic trauma from peaks) but interference with speech and warning signals (an accident risk), annoyance, sleep disturbance in nearby housing, and stress-related illness.

4. Control — source, path, receiver

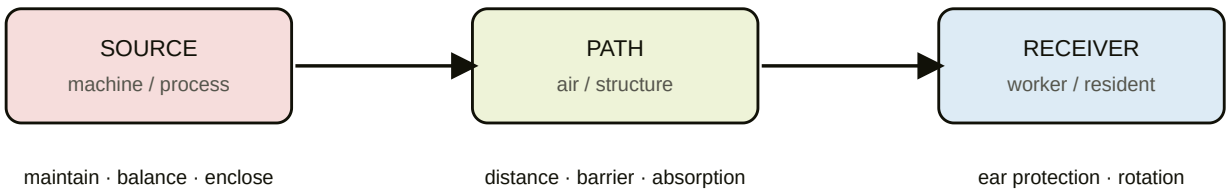


Figure 1 — The source–path–receiver model of noise control. Measures at the source protect everyone; measures at the receiver protect only the person wearing them, so the hierarchy runs left to right.

At the source (best): choose quieter machinery at purchase; maintain and balance rotating parts; reduce impact speeds; fit anti-vibration mounts to break structure-borne transmission; enclose the machine in a sealed, absorptively lined cabinet (a well-sealed enclosure commonly gives a large reduction; a badly sealed one gives little — the weakest opening sets the performance). **Along the path:** increase distance (6 dB per doubling in the open), erect barrier walls between plant and housing, line shed ceilings and walls with sound-absorbing material to cut the reverberant build-up, and plan layout so noisy plant sits far from boundaries and quiet offices. **At the receiver** (last resort): earplugs and earmuffs with realistic (not laboratory) attenuation assumed, job rotation to limit dose, quiet rest areas, and a hearing-conservation programme with periodic audiometry. A control plan states the expected reduction of each measure and verifies it by re-measurement — the same worked-example arithmetic, run in reverse.

Glossary

Decibel (dB) — logarithmic unit of sound level: 10 × $\log_{10}$ of the intensity ratio. **dB(A)** — A-weighted level, approximating the ear's sensitivity. **L_{eq}** — equivalent continuous level: the steady level carrying the same energy as a varying noise over the period. **Dosimeter** — wearable meter accumulating a worker's noise dose. **Octave-band analysis** — measurement of level by frequency band, used to design control. **Noise-induced hearing loss** — permanent threshold shift from chronic over-exposure. **Reverberant field** — sound built up by reflections inside a room. **Insertion loss** — reduction achieved by fitting a device such as an enclosure or barrier.

Key points

• Decibels add by intensities, not by numbers: two equal 88 dB sources give 91 dB, and the loudest source dominates any sum (90 + 88 + 84 = 92.7 dB).
• A point source loses 6 dB per doubling of distance in the open: 95 dB at 1 m becomes ≈ 76.9 dB at 8 m.
• Evaluation = survey (L_{eq}), dosimetry, octave bands, comparison with occupational criteria (≈ 85 dB(A)/8 h, 3 dB exchange) and zone-wise boundary limits.
• Control hierarchy: source first (maintenance, mounts, enclosures), then path (distance, barriers, absorption), receiver last (protection, rotation, audiometry).

Practice

1. Four identical machines each produce 85 dB(A) at a workstation. Find the combined level, and the level if one machine is switched off.
2. A compressor measures 92 dB(A) at 2 m in the open. Estimate the level at 16 m, and state the assumption behind the estimate.
3. A worker spends 4 h at 91 dB(A) and 4 h at 82 dB(A). Using the 3 dB exchange idea (85 dB(A) for 8 h = 100 % dose; each +3 dB halves the allowed time), estimate the worker's dose as a percentage.
4. List the steps of a noise evaluation survey for a weaving factory whose neighbours complain at night.
5. Why is an enclosure's real performance set by its worst-sealed opening? Name two other source controls and two path controls for a generator room.

Answer key

1. Four equal sources: 85 + 10 $\log_{10}$ 4 = 85 + 6.0 = **91.0 dB(A)**. Three sources: 85 + 10 $\log_{10}$ 3 = 85 + 4.8 = **89.8 dB(A)**.
2. Distance ratio 16/2 = 8 → fall = 20 $\log_{10}$ 8 = 18.1 dB → level ≈ 92 − 18.1 = **73.9 dB(A)**. Assumption: point source radiating into free field (no reflections); indoors or near reflecting surfaces

the level would be higher.

3. At 91 dB(A) the allowed time = 8 ÷ 2^((91−85)/3) = 8 ÷ 4 = 2 h; 4 h there = 200 % of that allowance… more precisely dose = (4/2) + (4/allowed at 82). At 82 dB(A) allowed time = 8 × 2^((85−82)/3) = 16 h; 4 h = 25 %. Total dose ≈ 200 % + 25 % = **≈ 225 %** — well over 100 %, so exposure must be reduced. (Accept ≈ 200–225 % with correct method.)
4. Steps: identify sources and operating pattern (including night running); measure L_{eq} at operator positions and at the boundary nearest the housing, day and night, with a calibrated sound level meter; dosimetry for representative workers; octave-band readings where control design is needed; compare with occupational criteria and with the zone limits for the boundary (mixed/industrial by day and night); report sources ranked by contribution.
5. Sound escapes through the path of least resistance: an unsealed gap passes sound almost unattenuated, so the total leakage is dominated by the largest opening, whatever the wall quality. Other source controls: anti-vibration mounts, maintenance/balancing (also: quieter replacement machine, exhaust silencer). Path controls: increased distance/re-siting, barrier wall, absorptive lining of the room (any two each).

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib